Developing Sustainable Procurement System Based on Enterprise Resource Planning

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Abstrak

Melihat persaingan yang semakin meningkat di industri manufaktur, penerapan Manajemen Rantai Pasok Berkelanjutan dapat menjadi solusi untuk meningkatkan efisiensi rantai pasok suatu perusahaan dan juga meningkatkan daya saing dengan para kompetitor. PT. Elco Indonesia Sejahtera merupakan salah satu perusahaan yang bergerak di bidang industri penyamakan kulit yang berlokasi di Garut, Jawa Barat. Perusahaan ini belum memiliki sistem untuk mendukung penerapan Manajemen Rantai Pasok Berkelanjutan dalam proses bisnis perusahaannya sampai saat ini, termasuk untuk bagian pengadaan. Selain itu, perusahaan juga belum memiliki system untuk memantau proses bisnisnya. Oleh karena itu, penelitian ini bertujuan untuk merancang sistem informasi terintegrasi untuk bagian pengadaan dengan menggunakan sistem yang berbasis Enterprise Resource Planning (ERP), serta melakukan perancangan dashboard monitoring agar proses bisnis dapat lebih efektif, efisien, dan tentunya mampu mengelola Manajemen Rantai Pasok Berkelanjutan dengan baik. Hasil dari penelitian ini berupa rancangan sistem dan juga dashboard monitoring yang berbasis ERP untuk mendukung penerapan Manajemen Rantai Pasok Berkelanjutan.

Kata kunci—Manajemen Rantai Pasok Berkelanjutan, Enterprise Resource Planning (ERP), Pengadaan Berkelanjutan

Abstract

Seeing the increasing competition in the manufacturing industry, the implementation of Sustainable Supply Chain Management (SSCM) can be a solution to improve the efficiency of a company's supply chain and also increase competitiveness with competitors. PT. Elco Indonesia Sejahtera is a company engaged in the leather tanning industry located in Garut, West Java. This company does not yet have a system to support the implementation of Sustainable Supply Chain Management in its business processes, including for the procurement department. In addition, the company also does not have a system to monitor its business processes. Therefore, this study aims to design an integrated information system for the procurement department using a system based on Enterprise Resource Planning (ERP), as well as to design a monitoring dashboard so that business processes can be more effective, efficient, and of course, able to manage SSCM well. The results of this study are in the form of a system design and a monitoring dashboard based on ERP to support the implementation of SSCM.

Keywords— Sustainable Supply Chain Management (SSCM), Enterprise Resource Planning (ERP), Sustainable Procurement

1. INTRODUCTION

The Central Statistics Agency recorded the growth of several manufacturing industry sectors in the second quarter of 2018. Production growth increased by 4.36% compared to the second quarter of 2017. The types of manufacturing that experienced the highest production growth were the leather and leather goods industry which reached 27, 73%. There are many leather-producing areas in Indonesia, one of which is in Garut Regency, West Java. The Garut Regency Industry, Trade, Cooperative and SME Service (2017) states that the high population and production capacity of livestock in Garut Regency initiated the growth and development of the leather tanning industry in Garut Regency. The impact that resulted from the economic side was very much felt by the existence of this leather tanning industry. But besides that, this also raises a negative issue because the waste produced can threaten the environmental ecosystem [1]. Therefore, the industry must implement Sustainable Supply Chain Management (SSCM) practices that help the industry reduce its environmental issues and increase its sustainable activities by considering aspects of green innovation [2]. Sustainable Supply Chain Management (SSCM) is the relationship between the supply chain where three economic, environmental and social factors are considered in its operations [3]. The environmental aspect of SSCM aims to minimize or eliminate waste in hazardous chemicals, emissions, energy and waste along the supply chain process [4].

SSCM involves entities from the procurement cycle to the production cycle to the distribution cycle with different interests but having the main objective of meeting the needs of the end consumers [5]. Therefore, procurement plays a vital role in SSCM activities in companies, such as what happened to PT Elco Indonesia Sejahtera, a leather tanning industry in Garut, West Java, Indonesia. The procurement department of PT Elco Indonesia Sejahtera has a significant role in supply chain activities because all raw material purchasing activities required for the production process are carried out in procurement. So that management and monitoring in procurement are needed to support the implementation of SSCM in the procurement section of PT Elco Indonesia Sejahtera. The indicator that can be used to apply SSCM to procurement is the sustainable procurement indicator. The Head of the Construction Development Agency of the Ministry of Public Works explained that sustainable procurement is a process of procuring goods that is carried out by assessing the entire construction process starting from planning, construction and service life (life cycle cost), there by providing economic and social benefits and reducing negative impacts to the environment.

Currently, PT Elco Indonesia Sejahtera does not have a system to support SSCM implementation. Therefore, to improve the company's performance, PT Elco Indonesia Sejahtera needs to develop an information system with integrated data so that business processes can be more effective, efficient, and of course, able to manage SSCM well. To solve these problems, we need a system that can integrate and monitor company business processes using an Enterprise Resource Planning (ERP) system. ERP is a system that can integrate all company supply chains and business processes by using a database in one place [6]. PT Elco Indonesia Sejahtera also does not have a monitoring system for the procurement business process, so this research will also design a monitoring dashboard. The dashboard is an information system application model to present performance quality information from a company or organizational institution [7]. This study aims to help the leather tanning industry to apply environmentally friendly criteria to create sustainable supply chain management in the procurement department using an ERP system.

2. METHODOLOGY

In implementing a sustainable procurement system, this study uses the QuickStart methodology. This methodology has three phases, as shown in Figure 1. The first phase is a kick-off call, which in this phase defines the company's business processes, especially in the procurement section. The second phase is analysis. At this stage, an analysis of the company's existing business processes will be carried out, then determine the target business processes that will be applied to the system. Then the third phase, namely Configuration, is the stage for configuring and adjusting the system.

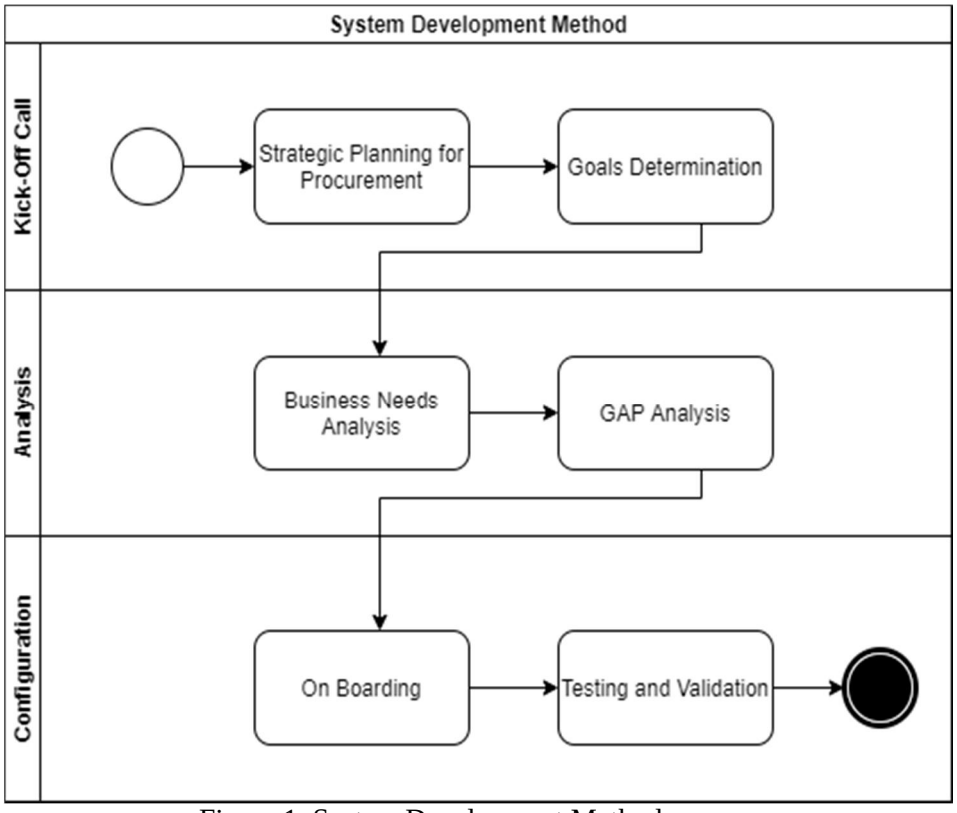


Figure 1. System Development Method

2.1 Enterprise Resource Planning (ERP)

Enterprise Resource Planning (ERP) is the software program used by companies to integrate and coordinate information on their business areas in real-time [8]. ERP aims to help manage every resource in the company. These resources include money, people, machines, time and materials [9]. In principle, with an ERP system, the industry or company can run optimally and can reduce inefficient operational costs, such as inventory costs and losses due to technical errors [6].

2.2 Sustainable Supply Chain Management

Sustainable Supply Chain Management (SSCM) is simply defined as the relationship between business supply chains where three economic, environmental and social factors are considered in its operational activities [3]. The environmental aspect of SSCM aims to minimize or eliminate waste in hazardous chemicals, emissions, energy and waste along the supply chain process where the supply chain process starts from product design, material selection,

manufacturing process, delivery of the final product to the customer and management of the end of life of the product [4].

2.3 Sustainable Procurement

Sustainable procurement is known as the environmental purchasing approach, which ensures that the items purchased are in line with the desired ecological attributes, such as reusable, recyclable, and contain non-toxic materials. Sustainable procurement is also related to material substitution through the right source of raw materials to reduce waste by minimizing hazardous waste, thus requiring good cooperation with suppliers to develop environmentally friendly products. Sustainable procurement focuses on cooperating with suppliers to create products that are environmentally sustainable [10].

2.4 Odoo

Odoo is the world's most accessible all-in-one management software. Odoo has several characteristics, such as customizable, comprehensive, modular and open-source [11]. Because it is an open-source application, users can freely customize it according to their individual preferences. Odoo's strengths include providing various business functions and an easy-to-understand user interface. Odoo has many modules that users can download and use immediately. Most of all, the essential modules can be downloaded for free. Some of these modules are manufacturing, inventory, sales, inventory, procurement, etc [12].

2.5 Dashboard

The dashboard is defined as a visual display of the most critical information needed to achieve one or more goals, which are combined and arranged in one screen so that they can be monitored at a glance. The dashboard provides a display interface with various forms such as diagrams, reports, visual indicators, and alert mechanisms combined with dynamic and relevant [13]. The concept of the performance dashboard has been around for many years, and the dashboard has been widely adopted by companies or businesses [11].

3. RESULT AND DISCUSSION

This study discusses problems regarding the procurement business process in the leather tanning industry, which in carrying out its business processes have not applied environmentally friendly indicators. In addition, business activities are still carried out manually, are not integrated and cannot be monitored in real-time. So it is necessary to implement an ERP-based system that can help automate and integrate business processes in the company. This research will design a sustainable procurement system based on ERP and also a system that can display data in the form of reports to simplify the process of monitoring company performance and performance in the form of a dashboard.

3.1 Kick-Off Call

This stage is the initial stage of the QuickStart method. At this stage, the business process of the procurement department will be determined. This stage aims to understand the company's business processes so that it can determine the system implementation plan.

Table 1. Strategic Planning

Environment	Development	Basic Concept
There is no particular system to manage the procurement business process in the leather tanning industry. The method of exchanging and inputting data is still done manually using physical documents. Its business activities also have not implemented environmentally friendly standardization so that it can trigger environmental pollution.	Designing an ERP system and monitoring dashboard in the procurement department by implementing environmentally friendly indicators	Enterprise Resource Planning
		Sustainable Procurement
		Key Performance Indicator
		Odoo
		Dashboard

According to Table I, it can be concluded that the company does not yet have an integrated system to support the procurement business process. In addition, the business process has not implemented environmentally friendly standardization. Therefore, an ERP system and monitoring dashboard will be designed by implementing environmentally friendly indicators. The existing basic concept is helpful for providing understanding for implementation.

3.2 Analysis

After determining the objectives to be achieved in accordance with the needs of the procurement department, at this stage, an analysis of the company's existing business processes will be carried out, then determine the target business processes that will be applied to the system.

3.2.1 Supplier Selection

1. *User Requirement*
   In the system, the purchasing department can submit a request for a quotation to find the best certified green supplier.
2. *Existing System (As-is)*
   There is no selection of suppliers and materials based on green indicators. The company still uses the commonly used supplier.
3. *Proposed System (To-be)*
   Supplier selection can be done easily because the system already contains supplier information according to company needs and has been certified.

In companies, there is no supplier selection business process. Then the system will apply the business process as shown in Figure 2.

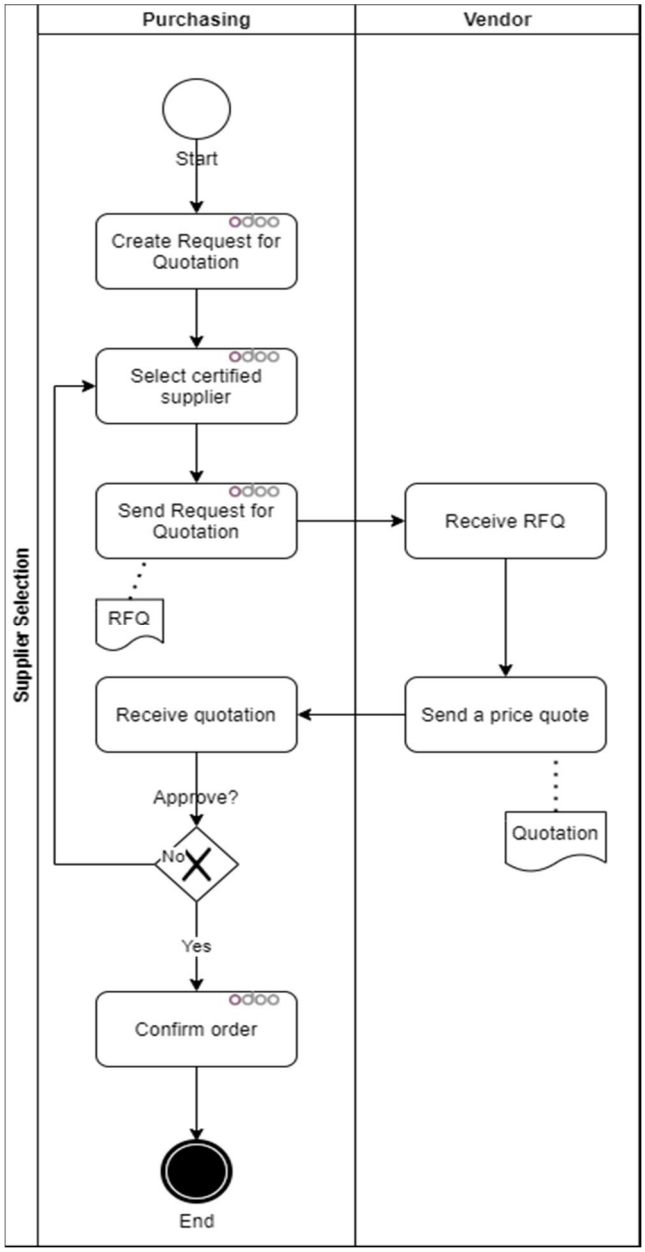


Figure 2. Supplier Selection Process Business Proposed

3.2.2 Purchase Order

1. *User Requirement*
   The system can make a purchase order according to the order list from the warehouse division.
2. *Existing System (As-is)*
   The purchasing department still uses physical documents to purchase materials.
3. *Proposed System (To-be)*
   There is a system containing information about the material to be purchased so that it helps the purchasing division to make material purchases.

In companies, the purchase order business process already exists but does not yet use the system. In Figure 4, it can be seen that the purchasing division can create purchase order documents through the system. So that business processes can run more efficiently.

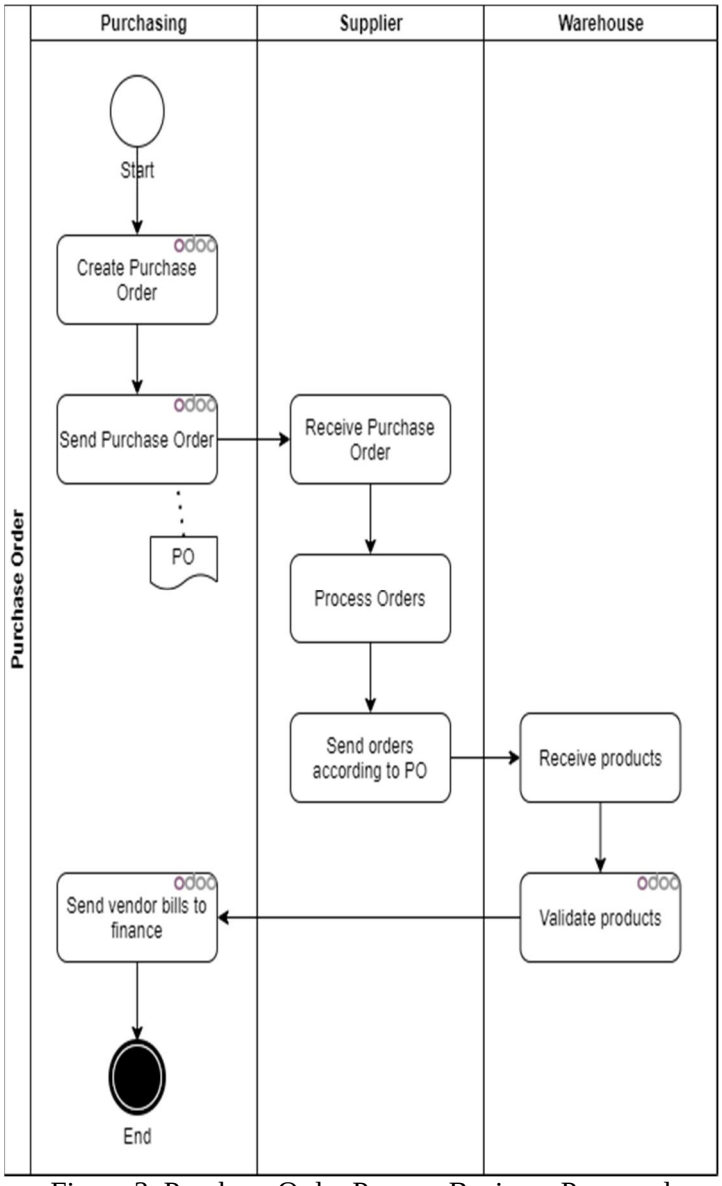


Figure 3. Purchase Order Process Business Proposed

3.3 Configuration

At this stage, the Odoo system configuration and customization will be carried out and the monitoring dashboard design based on the results of the new business process analysis. A Key Performance Indicator is needed to support the development of a sustainable procurement system in designing ERP-based applications. Key Performance Indicators serve to report the percentage of green materials and green suppliers in the system.

Table 2. Key Performance Indicator

KPI	Description
ISO 14000 Certified	Suppliers that have ISO 14000 environmental management system certification
Reuseable Material	Materials that can be reused for the production process
Recyclable Material	Material that can be recycled from the total waste
Material Biodegradable	The materials used can decompose in a short time
Materials Contaminated by Hazardous Chemicals	Material is contaminated (colour or smell)

Based on Table 2, there are five environmentally friendly attributes for green materials and green suppliers to help companies control the materials and suppliers used by the company to meet environmentally friendly standards.

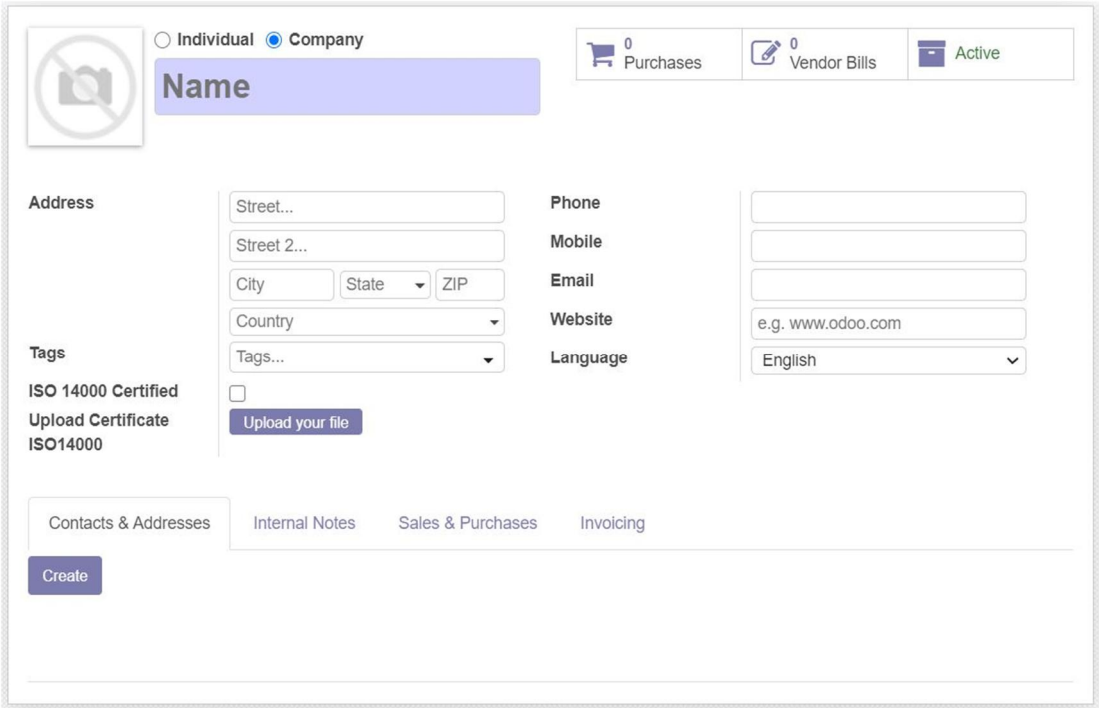


Figure 4. Supplier Data Feature

Based on the picture above, it can be seen that purchasing division staff can create supplier master data. There is a field that indicates that the supplier is ISO 14000 certified. Then a certificate file can be uploaded as proof that the supplier is ISO 14000 certified.

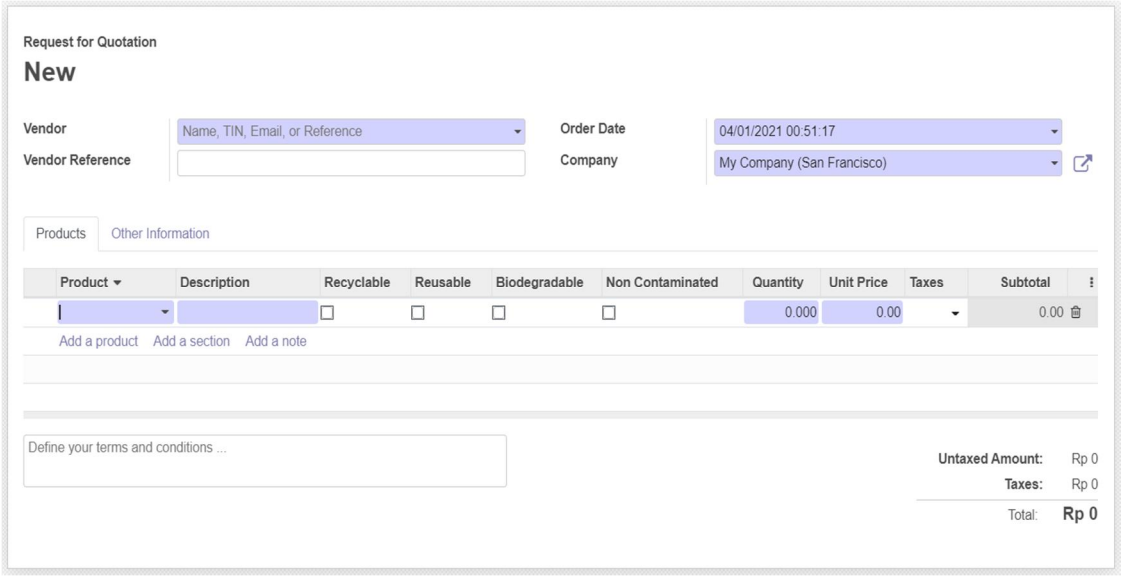


Figure 5. Request for Quotation Feature

In figure 5, the system will have a Request for Quotation (RFQ) form, which will be addressed to suppliers. There are several parameters to propose environmentally friendly materials. These parameters include Recyclable Materials, Reusable Materials, Biodegradable Materials, and Non-Contaminated Materials. Purchasing staff can determine the parameters desired by the supplier. If the staff proposes to purchase green materials, there will be an additional price. Later, the system will immediately calculate the total price according to the parameters proposed.



Figure 6. Procurement Dashboard

Figure 6 shows a visualization of the proposed procurement dashboard. Several data will be displayed, including supplier data, percentage of green material, percentage of dangerous materials, delivery time, supplier rating, cost of purchase orders, cost reduction and cost savings by year and supplier category. Supplier data displays information about the number of suppliers contained in the parent company data. Then it is classified into two groups, namely suppliers that are ISO 14000 certified and those that are not.

Then there is data that displays information about the percentage of green material. The higher the percentage indicates that the materials used by the company are environmentally friendly because these materials are reusable, recyclable, and biodegradable. So that the waste management process becomes more manageable. In addition, there is information on the percentage of dangerous materials. The lower the percentage indicates that the less use of materials contaminated with hazardous materials so that it has a good impact on the company.

Then there is information about the delivery time. This data is analyzed by comparing suppliers. The indicators used are early, on time, and late delivery. Based on Figure 6, it can be seen that supplier 4 has better performance than other suppliers because it is never late in making deliveries. Even though the percentage of supplier 1 is higher, there is still a percentage of late delivery.

There is also a supplier rating, which contains information about the three suppliers with the best ratings. Then there is information that displays all purchase costs with a five-year trend indicator. Then there are two other metrics that actually convey the same information but are expressed in two different ways: cost reduction and cost savings. This is analyzed based on five-year trends and by supplier category. It plays a key role in the procurement department, as it helps measure the large savings realized over time and helps in monitoring the financial performance of the purchasing function. Companies can use this figure because it has a direct impact on profits and losses.

4. CONCLUSION

Based on the results of the research that has been done, it can be concluded that the development of an ERP-based sustainable procurement system aims to create an environmentally friendly business process procurement information system. Companies can select suppliers that have been certified environmentally-friendly and choose the desired material criteria in the system. Then the existing data will be displayed in the form of a dashboard. This can help companies analyze current business processes and help make decisions. The result of this research is a system design for ERP-based procurement business processes for the needs of the leather tanning industry. The application of sustainable indicators can assist companies in creating an environmentally friendly industry.

5. SUGGESTION

Based on the research results, it is suggested that if there is further research on this industry, it is hoped that further researchers will be able to explore a broader range of Key Performance Indicators as environmentally friendly attributes.

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